

MATLAB: Session 3

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1 Theoretical Background

We now want to explicitly find solutions to linear systems. To do so, we perform the Gauss elimination. The material is taken from the main reference of the course, *Algebra lineare e geometria analitica* (A. Bernardi, A. Gimigliano). It is substantially the same exact content of pages 79-82, but translated into English.

1.1 Gaussian Elimination

We describe in detail the steps that make up the algorithm that explains how to perform Gaussian elimination on a $(m \times n)$ matrix A :

- i) Consider the first nonzero column of A ; let its index be j_1 . There are two possibilities: either in that column the first element (that is, a_{1j_1}) is nonzero, or it is not, and $a_{i_1j_1}$ is the first nonzero entry (with $i_1 > 1$). In the first case we move to step iii), in the second case to step ii).
- ii) Exchange the first and the i_1 -th row, obtaining a new matrix in which the element in position $(1, j_1)$ is nonzero.
- iii) Eliminate in the j_1 -th column all the nonzero entries from the second row downwards, leaving the first row unchanged and using the first row to carry out the elimination on the others, obtaining a new matrix such that $a_{ij_1} = 0$ for $i = 2, \dots, m$.
- iv) Now leaving the first row unchanged, repeat steps i), ii), iii) on the $(m-1) \times n$ matrix formed by the remaining $m-1$ rows, unless they are all zero, in which case we stop.

Let us now see what we obtain with this procedure. Starting from a matrix $A \in \mathbb{R}^{m,n}$, it will produce a matrix $A' = (a'_{ij}) \in \mathbb{R}^{m,n}$, which will surely be upper triangular (it has only zeros below the diagonal, that is, in the positions

a'_{ij} with $i > j$). But the effect of Gaussian reduction also gives something more: it brings the matrix into a *row echelon form*, that is, of the type

$$\begin{bmatrix} 0 & \cdots & 0 & \underline{p_1} & * & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \underline{p_2} & * & \cdots & \cdots & \cdots & \cdots & * \\ 0 & \cdots & \cdots & 0 & \cdots & 0 & 0 & \underline{p_3} & * & \cdots & \cdots & * \\ \vdots & & & & \ddots & & & & \ddots & & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & \underline{p_r} & * & * \\ 0 & \cdots & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}$$

where, using the notations of ii), $p_1 = a'_{1j_1}$.

Our matrix is not only upper triangular, but also has the property that below the first nonzero element of each row, there appear only zeros (this is what we mean by a row echelon matrix).

We have denoted by $p_1, p_2, \dots, p_r$ the first nonzero elements of each row of a row echelon matrix; they are called *pivots*. Note that the notion of pivot makes sense only for row echelon matrices, and that if r is the number of pivots (that is, the number of nonzero rows), then $r \leq m$, and that if $r = m$, then there are no final zero rows (as indicated in the matrix above). We also have $r \leq n$, since the p_i occupy distinct columns, so they cannot exceed the number of columns of A .

Definition. Let $A \in \mathbb{R}^{m,n}$, and A' the matrix obtained from A by Gaussian elimination. The number of pivots appearing in the matrix A' is called the *rank* of the matrix A , and is denoted by $r = r(A)$.

Remark. We have $r(A) = 0$ if and only if A' is the zero matrix.

1.2 Solving Linear Systems via Gaussian Elimination

Consider the $(m \times n)$ linear system $Ax = b$. We now show how it is possible to determine whether a system has solutions and how many, by analyzing the matrix $(A|b)$. Consider the system:

$$\begin{cases} x - y + z = 3 \\ x + y - 2z = 4 \\ 3x - y = 5 \end{cases}$$

Applying Gaussian elimination, we obtain:

$$\begin{cases} x - y + z = 3 \\ 2y - 3z = 1 \\ 2y - 3z = -4 \end{cases} \implies \begin{cases} x - y + z = 3 \\ 2y - 3z = 1 \\ 0 = -5 \end{cases}$$

thus the system is inconsistent. Let us see the same thing by operating on the matrix $(A|b)$:

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 1 & 1 & -2 & 4 \\ 3 & -1 & 0 & 5 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 2 & -3 & 1 \\ 0 & 2 & -3 & -4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & 2 & -3 & 1 \\ 0 & 0 & 0 & -5 \end{array} \right].$$

So, if the system is inconsistent, Gaussian elimination yields an equation (after eliminating variables) of the type $0 = -5$; that is, during elimination of $(A|b)$ we find a row $(0 \ 0 \ \dots \ 0 \ 0 \ p)$ with $p \neq 0$, that is, a pivot in the last column, as in the previous example where the nonzero pivots are 1, 2, -5 , and the -5 lies in the last column.

THE SYSTEM HAS NO SOLUTIONS

if and only if

in the Gaussian reduction of $(A|b)$ there is a pivot in the last column.

What does this mean in terms of the rank of the matrices involved? If $(A|b)$ has rank r , then A will have rank $r - 1$, because among the r pivots of $(A'|b')$, the last one lies on B' , and thus A has $r - 1$ pivots. Hence we can rephrase the previous statement as follows:

THE SYSTEM HAS NO SOLUTIONS

if and only if

$$r(A) \neq r(A|B).$$

And if $r(A) = r(A|B) = r$? Then the pivots of $(A'|B')$ are all contained in the part A' , and in this case the Gaussian elimination brings $(A'|B')$ to the form:

$$\left[\begin{array}{cccccccccccc} 0 & \dots & 0 & \underline{p_1} & * & \dots & \dots & \dots & \dots & \dots & \dots & * \\ 0 & \dots & 0 & 0 & 0 & \underline{p_2} & * & \dots & \dots & \dots & \dots & * \\ 0 & \dots & \dots & 0 & \dots & 0 & 0 & \underline{p_3} & * & \dots & \dots & * \\ \vdots & & & & \ddots & & & & \ddots & & & \vdots \\ 0 & \dots & 0 & \dots & 0 & 0 & 0 & 0 & \dots & \underline{p_r} & * & * \\ 0 & \dots & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 & 0 & 0 \end{array} \right]$$

Each pivot $p_1, \dots, p_r$ is the coefficient of an unknown and allows us to express that unknown as a function of the others (in the equation corresponding to that row). Therefore, in total, there remain $n - r$ free variables; that is, the system has ∞^{n-r} solutions (note that if $n = r$, this means that we have zero free variables, i.e., the system has a unique solution).

Summing up, we have proved the following theorem, known as the *Rouché–Capelli theorem*:

Theorem 4.3.1 (*Rouché–Capelli Theorem*). The linear system $A \cdot X = B$ has solutions if and only if

$$r(A) = r(A|B).$$

In this case, the solutions are ∞^{n-r} (that is, there are $n - r$ free variables, and there is a unique solution if $n = r$).

1.3 Matlab Implementation

Consider the linear system

$$\begin{cases} x_1 + x_2 - 2x_3 = 3 \\ 2x_1 + x_2 - 3x_3 - x_4 = 5 \\ 3x_1 + 2x_2 - 5x_3 - x_4 = 8 \\ x_1 - x_2 - 2x_4 = 1 \end{cases}.$$

Define the coefficient matrix and the vector of constants

$$A = \begin{bmatrix} 1 & 1 & -2 & 0 \\ 2 & 1 & -3 & -1 \\ 3 & 2 & -5 & -1 \\ 1 & -1 & 0 & -2 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 5 \\ 8 \\ 1 \end{bmatrix}.$$

Apply Gaussian elimination to the augmented matrix $(A|b)$:

$$\begin{aligned} \left[\begin{array}{cccc|c} 1 & 1 & -2 & 0 & 3 \\ 2 & 1 & -3 & -1 & 5 \\ 3 & 2 & -5 & -1 & 8 \\ 1 & -1 & 0 & -2 & 1 \end{array} \right] & \longrightarrow & \left[\begin{array}{cccc|c} 1 & 1 & -2 & 0 & 3 \\ 0 & -1 & 1 & -1 & -1 \\ 0 & -1 & 1 & -1 & -1 \\ 0 & -2 & 2 & -2 & -2 \end{array} \right] \\ & \longrightarrow & \left[\begin{array}{cccc|c} 1 & 1 & -2 & 0 & 3 \\ 0 & -1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

Hence $\text{rk}(A) = \text{rk}(A|b) = 2 < n = 4$. There are $n - \text{rk}(A) = 2$ free variables. Let $t = x_3$ and $h = x_4$. From the last echelon matrix, we get

$$\begin{cases} x_1 + x_2 - 2x_3 = 2, \\ -x_2 + x_3 - x_4 = -1, \end{cases} \implies (x_1, x_2, x_3, x_4) = (2+t+h, 1+t-h, t, h), \quad \forall t, h \in \mathbb{R}.$$

Step-by-step MATLAB guide

1. Set A and b as above and form $Ab = [A \ b]$.
2. Compute $\text{rk}A = \text{rank}(A)$ and $\text{rk}Ab = \text{rank}(Ab)$. Since $\text{rk}A == \text{rk}Ab < 4$, the system has infinitely many solutions, precisely $\infty^{4-2=2}$.
3. Perform the following row operations on Ab :

$$\begin{aligned} R_2 &\leftarrow R_2 - 2R_1, & R_3 &\leftarrow R_3 - 3R_1, & R_4 &\leftarrow R_4 - R_1, \\ R_3 &\leftarrow R_3 - R_2, & R_4 &\leftarrow R_4 - 2R_2. \end{aligned}$$

4. Read off the parametric solution $(x_1, x_2, x_3, x_4) = (2+t+h, 1+t-h, t, h)$.
5. Verify that $A \cdot \text{sol} = b$.

MATLAB code

```
% Coefficients and constants
A = [1  1 -2  0;
      2  1 -3 -1;
      3  2 -5 -1;
      1 -1  0 -2];
b = [3; 5; 8; 1];

% Augmented matrix and ranks
Ab = [A b];
rkA = rank(A);
rkAb = rank(Ab);
hasSolutions = (rkA == rkAb);    % true; rk = 2 < n = 4, 2 param.

% Gaussian elimination
Ab(2,:) = Ab(2,:) - 2*Ab(1,:);
Ab(3,:) = Ab(3,:) - 3*Ab(1,:);
Ab(4,:) = Ab(4,:) - Ab(1,:);
Ab(3,:) = Ab(3,:) - Ab(2,:);
Ab(4,:) = Ab(4,:) - 2*Ab(2,:);

% Parametric solution
syms t h real
sol = [2 + t + h; 1 + t - h; t; h];

% Check A*sol = b
Asol = A*sol;
simplify(Asol - b)    % zero vector, as expected
```

2 Exercises

Part A – Symbolic computations

A.1 Create a symbolic variable **x**. Define the expression

$$\text{exp} = x^4 + 2x^3 - 15x^2 + 6x^3 + 12x^2 - 90x - 7x^2 - 14x + 105.$$

Assign it to the variable **exp**. Substitute $x = 2$ into **exp** and save the result in **exp2**.

A.2 Factorize *exp* using the command **factor**, saving the result in the variable **fatt**. Simplify *exp* using **simplify**, saving the result in the variable **simpl**. Observe the difference between the two outputs.

A.3 Define the equation

$$x^4 + 8x^3 - 10x^2 - 104x + 105 = 0.$$

Save it in the variable **eq**. Solve **eq** with respect to **x** and save the result in the variable **solx**. Comment on the form of the output.

A.4 Define the symbolic variables **a**, **b**, **c**. Define the expressions

$$\text{eqn}_1 = 5a + b + 2c, \quad \text{eqn}_2 = 7b - 3c.$$

Assign them to the variables **eqn1** and **eqn2** respectively.

A.5 Define the equation

$$\text{eqn}_2 = \text{eqn}_1.$$

Save it in the variable **eqn3**.

A.6 Solve **eqn3** with respect to **a** and store the solution in the variable **sol1**. Then substitute $a \mapsto 4a + 2$ directly into **eqn3** and solve again with respect to **a**, saving the new solution in the variable **sol2**.

Part B – Compatibility of Linear Systems

B.1 Consider the linear systems

$$\begin{array}{ll} A) \begin{cases} x + 2y - z = 1, \\ 2x + 4y - 2z = 2, \\ -x - 2y + z = -1. \end{cases} & B) \begin{cases} x - y + z = 3, \\ 2x + y + z = 1, \\ 3x + 2y + 2z = 4. \end{cases} \\ C) \begin{cases} x + y + z = 2, \\ 2x - y + 3z = 5, \\ x + 2y + 4z = 7, \\ 3x + y + 6z = 11. \end{cases} & D) \begin{cases} x + y + z = 1, \\ x + 2y + 3z = 2, \\ 2x + 3y + 4z = 2. \end{cases} \end{array}$$

Check in MATLAB whether each system is compatible. If so, determine whether the solution is unique or there are infinitely many solutions. In the latter case, specify the number of free parameters in the solution.

B.2 Consider the parametric linear systems, for $a \in \mathbb{R}$:

$$\begin{cases} x + y + z = 1, \\ x + ay + z = 2, \\ 2x + (a + 1)y + 2z = 3. \end{cases}$$

Check in MATLAB for which values of a the systems is compatible. Among such values of a , find those for which the solution is unique and those for which there are infinitely many solution. In the latter case, specify the number of free parameters in the solution.

B.3 Consider the parametric linear systems, for $k \in \mathbb{R}$:

$$\begin{cases} x + y + z = 1, \\ x + ky + z = 2, \\ 2x + (k + 1)y + 3z = 3. \end{cases}$$

Check in MATLAB for which values of a the systems is compatible. Among such values of a , find those for which the solution is unique and those for which there are infinitely many solution. In the latter case, specify the number of free parameters in the solution.

Part C – Solving Linear Systems: Gaussian Elimination

C.1 Solve system B) in **B.1**, performing Gaussian elimination on MATLAB.

C.2 Solve system A) in **B.1**, performing Gaussian elimination on MATLAB.

C.3 Solve the following system

$$\begin{cases} 3x + 2y - z = 1 \\ 7y + 2z = 1 \\ 6x + 2y = 0 \end{cases},$$

performing Gaussian elimination on MATLAB.

C.4 Solve the system in **B.2**, performing Gaussian elimination on MATLAB.

C.5 Solve the system in **B.3**, performing Gaussian elimination on MATLAB.